

INFRASTRUCTURE SPECIFICATION

64× NVIDIA B200
AI Inference Cluster

Production Research Platform Specification

Status	Draft — for architecture review
Purpose	Interactive multi-trillion MoE inference
Scale	1,000 registered users · 50 concurrent
Retention	All inference retained 1 year
Fabric	Arista 7060DX5 (data) · 7010TX-48 (OOB)
Orchestration	Kubernetes + NVIDIA GPU Operator
Revision	0.6 · 2026-07-18

Not a hyperscale training cluster.
Manageable enterprise research inference platform.

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1 Design Goals

Priority	Goal
1	Interactive inference latency
2	Large model support (multi-trillion MoE)
3	Research workload flexibility
4	High availability
5	Expandability

This is **not** a maximum-throughput hyperscale training cluster. It is a manageable enterprise research inference platform.

2 Physical Summary

Item	Specification
GPUs	64× NVIDIA B200
GPU servers	8
GPUs per server	8
Racks	4 (2×GPU + BOOT + STOR)
Rack density	32 GPUs/rack
Fabric	400GbE RoCEv2 rail-opt (Arista 7060DX5)
Host NICs	8×400GbE per server (1:1 GPU)
OOB	Arista 7010TX-48
Orchestration	Kubernetes + NVIDIA GPU Operator
Primary workload	MoE LLM inference
Inference archive	1-year object lake (≥750 TB usable)

3 Facility / Rack Zones

Rack	Role	Day-1
GPU-1	4× B200 (32 GPU) + rail leaves + OOB	Required
GPU-2	4× B200 (32 GPU) + rail leaves + OOB	Required
BOOT	K8s CP, PXE/registry, spines, NMS, brokers	Required
STOR	Hot model FS + 1-year object lake	Required (one)
STOR-2	Archive growth / isolation	Optional

BOOT: separate blast radius from GPU kW; holds DC bootstrap.

STOR: one rack is enough for 150–300 TB hot + ≥ 750 TB archive; add STOR-2 only past 1.5–2 PB usable or for rack-level Failure domains.

4 Compute Nodes

4.1 GPU Server Quantity

8 servers. Example platforms: NVIDIA MGX B200, Dell XE9680-class, HPE Cray XD, Supermicro HGX/B200, Lenovo ThinkSystem AI.

4.2 Per-Server Specification

Component	Specification
GPU	8× NVIDIA B200
GPU memory	1.4 TB HBM3e
GPU compute	1.4 PFLOPS FP8 (approx.)
CPU	2× AMD EPYC Turin / Genoa
CPU cores	128–192
System RAM	2 TB DDR5 ECC
Boot	2× 1.92 TB NVMe RAID1
Local NVMe	8× 3.84 TB
Data network	8× 400GbE (1 NIC per GPU; rail-optimized)
NIC class	ConnectX-7 / BlueField-3 SuperNIC class
Management	1× 1/10/25GbE
Power	8–10 kW

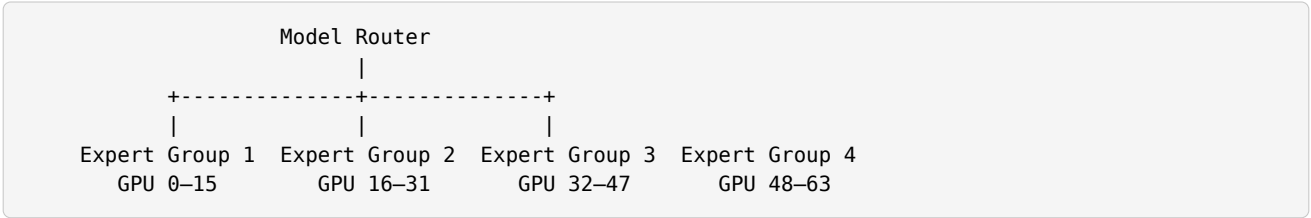
4.3 Cluster Aggregate Capacity

Resource	Amount
GPUs	64
HBM3e memory	11.5 TB
CPU cores	1,200
System RAM	16 TB
Local NVMe	250 TB
GPU FP8 compute	11–12 PFLOPS

5 GPU Parallelism Model

For a 3T MoE model, do **not** run one giant tensor-parallel job across all GPUs by default.

5.1 Logical Layout



5.2 Parallelism Dimensions

Dim	Role
TP	Tensor parallel (within node / NVLink domain)
EP	Expert parallel (across nodes over fabric)
PP	Pipeline parallel (optional depth split)

5.3 Example Configurations

TP=8, EP=8, PP=1
TP=4, EP=16, PP=1

Exact values depend on model architecture, expert count, and latency targets.

6 Network Architecture

Host NIC baseline

Each B200 has a **dedicated 400GbE** NIC (not 2 shared NICs per server).

Per server **3.2 Tbps**; cluster host edge **25.6 Tbps** (64×400G). Rail-optimized: 8 rail leaves.

For MoE inference, the fabric is part of the computer.

Standard platforms: Arista **7060DX5** (data plane, EOS, optional CloudVision); Arista **7010TX-48** (OOB).

6.1 Fabric Requirements

Feature	Requirement
Platform	Arista 7060DX5
NOS	Arista EOS
Link speed	400GbE (QSFP-DD)
Transport	RoCEv2
Congestion	ECN, PFC, DCQCN / advanced congestion mgmt
Load balancing	ECMP + Dynamic Load Balancing (DLB)
Overlay / control	BGP EVPN (VXLAN as needed)
Automation	eAPI / CloudVision
Tuning	AI / lossless Ethernet profile

6.2 Topology Rules

Important

RoCEv2 lossless Ethernet does not require equal leaf and spine counts.

What keeps the fabric well-behaved for MoE/EP traffic:

Requirement	Why it matters
Low/zero oversub leaf→spine	Avoid standing congestion that PFC turns into fabric-wide pause
PFC + ECN + DCQCN	Lossless queue without pause storms
Enough ECMP / DLB paths	Spread elephant flows; reduce incast hotspots
End-to-end jumbo MTU	Fewer packets, lower PPS stress
Buffer + queue telemetry	Catch microbursts before drops/pauses

Equal leaf/spine counts appear in some designs as a **square Clos with one SKU** — a convenience choice, not a lossless protocol rule.

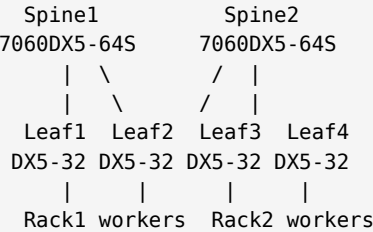
Spine count math:

$$\text{spine_ports_needed} \approx (\text{leaf_count} \times \text{uplinks_per_leaf}) / \text{ports_per_spine_toward_leaves}$$

For 1:1 fabric:
 $\sum \text{leaf uplink BW} \geq \sum \text{leaf downlink (host) BW}$

6.3 Topology Options

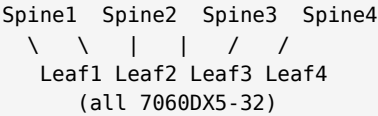
6.3.1 Option A — Efficient (default): 4 leaf + 2 spine



Role	Model	Qty
Rail leaf	DCS-7060DX5-32	8
Spine	DCS-7060DX5-64S (or 64E)	2

Host edge: 8 servers × 8×400G = **25.6 Tbps**. Rail model: one leaf per GPU index; GPU-*i* on every node → Rail-*i* leaf. **Per rail leaf 1:1**: 8×400G hosts → ≥8×400G toward spines.

6.3.2 Option B — Square Clos, one SKU: 4 leaf + 4 spine



Role	Model	Qty
Leaf	DCS-7060DX5-32	4
Spine	DCS-7060DX5-32	4

Use for identical BOM, full bipartite wiring, simpler sparing. **Not required for lossless.**

6.3.3 Chosen baseline

Default: 8× DX5-32 rail leaf + 2× DX5-64S spine (1 NIC per GPU, rail-optimized).

Rack	Leaves	GPU servers
Rack 1	Leaf1, Leaf2	worker01–04
Rack 2	Leaf3, Leaf4	worker05–08

6.4 Leaf Switches — Arista 7060DX5-32

Quantity: 8 rail leaves · **SKU:** DCS-7060DX5-32

Feature	Value
Ports	32× 400G QSFP-DD
Breakout	up to 128× 100G
L2/L3 throughput	12.8 Tbps
Forwarding	5.3 Bpps
Packet buffer	57 MB shared
Latency	from 850 ns
Typical power	289 W
Airflow	F-R or R-F (match rack)
HA hardware	1+1 PSU, N+1 fans
Routing	BGP EVPN, 128-way ECMP
AI features	DLB, advanced congestion, LANZ
QoS	PFC / ECN for RoCEv2

6.5 Spine Switches

6.5.1 Option A spines — 7060DX5-64 (qty 2)

Opt	Model	Ports	Tput	RU
A1	DCS-7060DX5-64S	64× 400G QSFP-DD	25.6 Tbps	2
A2	DCS-7060DX5-64E	32×800G → 64×400G	25.6 Tbps	1

Feature	Value
ECMP	up to 128-way
Latency	from 850 ns
Typical power	489 W (64S) / 548 W (64E)
Airflow	Front-to-rear
HA hardware	1+1 PSU, N+1 fans

6.5.2 Option B spines — 7060DX5-32 (qty 4)

Same SKU as leaves. Full bipartite; still size **uplink BW ≥ downlink BW** per leaf.

Default procurement: 8× DX5-32 rail leaf + 2× DX5-64S spine.

6.6 Server Connectivity

HGX B200 Server
GPU0..GPU7 — NIC0..NIC7 — 400G — Rail0..Rail7 leaves

Scope	Bandwidth
Per server	3.2 Tbps (8×400G)
Per GPU / NIC	400 Gbps

Scope	Bandwidth
Cluster aggregate server edge	25.6 Tbps (64×400 Gbps)

6.7 Port Plan (Planner Baseline)

Per leaf (7060DX5-32) — rack of 4 GPU nodes, dual-homed:

Use	Ports	Speed
GPU server downlinks	4	400G
Spine uplinks (2 per spine)	4–8	400G
Storage / services	2–4	400G
Spare / growth	remainder	400G

Target oversubscription leaf→spine: $\leq 1:1$ host-facing (prefer 2:1 uplink:downlink capacity where ports allow).

Per spine (7060DX5-64S):

Use	Ports	Speed
Leaf uplinks (4 leaves $\times$ 2–4)	8–16	400G
Storage cluster	2–4	400G
Border / DCI / services	2–4	400G
Spare / expansion rack	remainder	400G

6.8 Optics and Cabling

Link	Preferred media
Server $\leftrightarrow$ leaf (same rack)	400G-DR4 / DAC or AOC as length allows
Leaf $\leftrightarrow$ spine	400G-DR4 or FR4 per plant standards
Connector	QSFP-DD on 7060DX5-32 / 64S
Spares	20% optics/cables minimum

6.9 RoCEv2 / AI Ethernet Profile (Arista EOS)

Control	Setting direction
Priority flow control	Lossless queue for RoCE PFC priority
ECN	Enabled on RoCE queue; mark before drop
DCQCN	Host CNP reaction; switch ECN thresholds tuned
DLB	Prefer DLB over static hash for large flows
MTU	Jumbo (e.g. 9000) end-to-end
ECMP	Max-path $\geq$ leaf uplink count; resilient hashing
Telemetry	LANZ / queue depth; drop & ECN counters scraped

6.10 Out-of-Band / Management Network

Separate from the 400G RoCE fabric. **Never** carry GPU data-plane traffic.

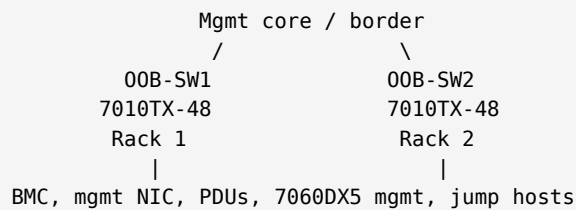
Standard platform: Arista **7010TX-48** (7010X series), EOS.

6.10.1 Platform

Item	Spec
Model	DCS-7010TX-48 (AC) or DCS-7010TX-48-DC
Form factor	1RU
Access ports	48× 10/100/1000BASE-T (RJ45)
Uplinks	4× SFP28 (1/10/25GbE)
Throughput	296 Gbps / 220 Mpps
Buffer	4 MB shared
Power	1+1 redundant PSU
Fans	1+1 hot-swap; reversible airflow
NOS	Arista EOS

6.10.2 Quantity and placement

Role	Qty	Placement
OOB leaf	2	1 per GPU rack (baseline)
Optional 3rd	1	Management rack if port pressure



6.10.3 Attachments

Endpoint	Speed	Notes
GPU server BMC	1G	Always-on lights-out
GPU OS mgmt NIC (opt)	1G	Rescue / PXE
K8s control-plane mgmt	1G	Not RoCE NICs
7060DX5 Management1	1G	EOS OOB mgmt
Storage / PDU / console	1G	As present
CloudVision / jump / IPMI	1G	Ops plane

OOB is **loss-tolerant best-effort** (no RoCE PFC).

Control	Requirement
Isolation	Dedicated VRF/VLAN; no route to RoCE data except controlled jump
Access	MFA jump host; no public BMC
ACLs	Mgmt subnets only → BMC/IPMI/SSH/HTTPS

Control	Requirement
Logging	EOS AAA + syslog/telemetry

7 Storage

Inference is **not** a training I/O path. Storage must load multi-TB MoE checkpoints, hold versions, archive all inference for research, and stay off the critical token path.

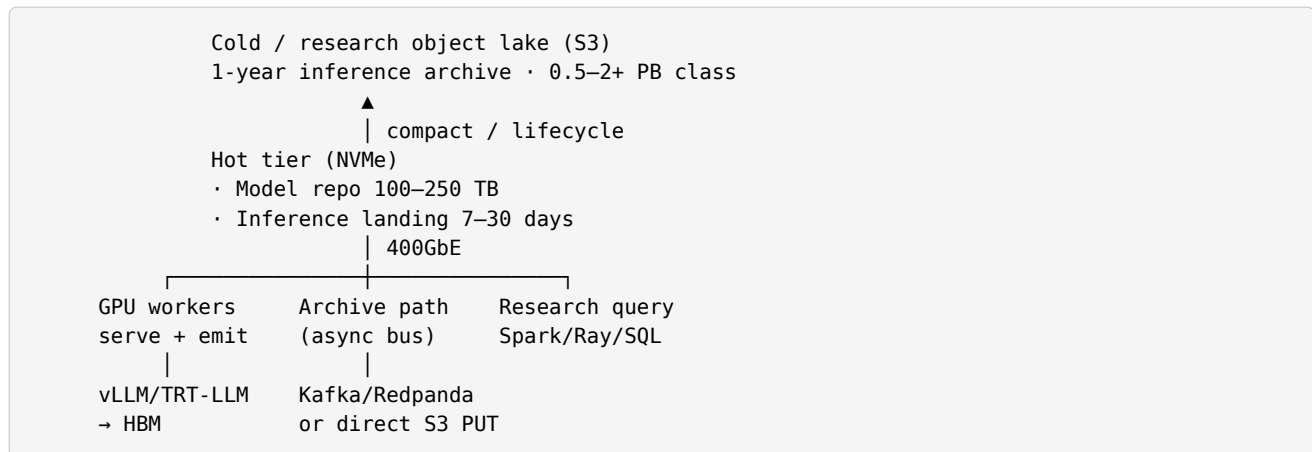
7.1 Role of Storage

Path	When	BW need	Latency
Model cold start	Deploy/restart	High burst	Startup only
Steady inference	Weights in HBM	0 for weights	GPU HBM
Inference archive	Every request	Sustained write	Must not block TTFT
KV spill (opt)	Long context	Med, local-first	Moderate
Datasets / eval	Ray jobs	Med sequential	Low-med
Ops metrics	Always	Low	Low

Design principles

1. Weights on hot shared NVMe; steady-state path is GPU HBM.
2. **All inference** (prompts, completions, metadata) retained ≥ 1 year.
3. Archive write is **async** — never on the critical token path.

7.2 Logical Architecture



7.3 Network Integration

Storage attaches to the **same Arista 7060DX5 data fabric** (not 7010TX-48 OOB).

Item	Spec
Fabric	7060DX5 leaf-spine
Host path	GPU NICs (data plane), not BMC/OOB
Isolation	Storage VRF/VLAN; QoS separate from RoCE EP if needed

Item	Spec
Transport	NFS/RDMA, parallel FS client, or S3 — product-dependent

BMC / storage controllers → 7010TX-48 OOB only.

7.4 Capacity and Performance Targets

7.4.1 Hot tier (models + short landing)

Metric	Target
Usable NVMe	100–250 TB models + 20–50 TB landing
Single MoE checkpoint	5–40 TB compressed/sharded per version
Concurrent versions	3–5 online
Aggregate model read	≥ 40 GB/s (stretch 80+)
Single-node model read	≥ 10–20 GB/s

7.4.2 Inference archive tier (1-year) — required

Policy: retain **every** inference exchange for **365 days**. Deletes only via explicit policy after retention.

Minimum fields per request:

Field	Required
request_id, trace_id, timestamp (UTC)	Yes
Tenant / user id (pseudonymized if needed)	Yes
Model id + version + serving revision	Yes
Full prompt / messages (as served)	Yes
Full completion tokens (as returned)	Yes
Sampling params (temp, top_p, max_tokens, seed)	Yes
Token counts, TTFT, TPOT, status, error	Yes
Router / expert stats	Research-preferred
Logprobs / top-k	Opt-in (large)
Tool calls / RAG citations	When used
Content hash + schema version	Yes

7.4.3 Capacity planning model

Assumptions: 50 concurrent, 32k context class, research use.

Scenario	What is stored	Sustained	1-year re-tained
Lean	Text + metadata; delta turns; parquet/zstd	0.5–2 TB/d raw	50–150 TB
Standard (default)	Full prompt+completion each call	2–8 TB/d raw	200–600 TB

Scenario	What is stored	Sustained	1-year re-tained
Full fidelity	1. logprobs / 32k-heavy	10–40 TB/d raw	1–3 PB

Procurement baseline

Size object/archive lake for **≥ 750 TB usable** (standard research + headroom), growth path to **1.5–2 PB**. Revisit after 30–90 days of measured bytes/request.

Peak 1k tok/s is **not** the size driver — **full prompt context logged per turn** is.

7.4.4 Archive performance

Metric	Target
Durability	Erasure-coded / multi-copy object; no single-disk year loss
Write path	Async; p99 archive lag ≤ 60s normal
Impact on TTFT	None
Query	Partition day/model/tenant; 7d interactive; year batch OK
Immutability	Append-only prefixes + lifecycle; WORM optional

7.5 Node-Local Storage

Each GPU server: **8× 3.84 TB NVMe** (30 TB raw) for kubelet/containerd, scratch cache, optional KV spill. **Not** primary model catalog.

7.6 Kubernetes Integration

Mechanism	Use
CSI driver	RWX for weights + datasets
PVC <code>models-rwx</code>	Read-mostly on GPU pods
initContainer / Job	Stage weights → local NVMe
KServe StorageUri	<code>pvc://</code> or <code>s3://</code>
Inference logger	Gateway plugin → bus/object
Images	Do not bake multi-TB weights into images

```

KServe InferenceService
  storageUri: pvc://models-rwx/moe-3t/v.../
  |
  ▼
GPU pod CSI mount → load to HBM
  ├── tokens to client
  └── async InferenceRecord → archive pipeline

```

7.7 Inference Capture and 1-Year Archive Pipeline

Requirement: 100% of inference traffic retained 1 year.

```

Client → API Gateway (redaction/ToS)
      → Model Server (vLLM / TRT-LLM / KServe)
          ├── response (sync)
          └── InferenceRecord (async)
              → buffer (NVMe ring / NATS / Redpanda / Kafka)
              → compactor workers
              → s3://inference-archive/year=.../month=.../day=.../
              → Ray / Spark / DuckDB / warehouse

```

Stage	Tech direction	Notes
Emit	OTel gen-AI or JSON schema	Stable <code>schema_version</code>
Transport	Kafka/Redpanda/NATS or S3	Bus if multi-consumer
Format	Parquet (zstd) primary	Year-scale scans
Partition	day / model / tenant	Prune queries
Index	Catalog DB request_id→key	Point lookup
Retention	365 days lifecycle	Legal hold available
Access	Research read IAM; write-only serve key	Segregate creds

Prefer **API gateway or dedicated logger** once (avoid TP/EP duplicate records). Dedupe on `request_id` if engine-side. PII hooks before durable write; policy denials → tombstone + reason.

7.8 Data Layout

Models (hot FS):

```

/models/moe-3t/v.../{config,shards,CHECKSUMS}
/embeddings/ /adapters/ /datasets/ /eval/ /artifacts/

```

Inference archive (object):

```

s3://inference-archive/
  year=2026/month=07/day=18/model=moe-3t/ver=.../part-*.parquet
  _catalog/  _schema/v1.json

```

7.9 Platform Options

Tier	Preferred options	Role
Hot models	Weka / BeeGFS / Lustre / Flash-Blade NFS	Checkpoint RWX
Inference archive	Ceph RGW, MinIO, Vast, Pure, ...	1-year lake
Bus (opt)	Redpanda / Kafka	Fan-out
Catalog	Postgres / OpenSearch	request_id lookup

Need	Choice
Model load	Parallel FS or appliance NFS on NVMe (100–250 TB)

Need	Choice
1-year inference	S3 API object lake $\geq$ 750 TB , EC, lifecycle 365d
Research query	Ray / external tables over parquet

7.10 Hardware Footprint (indicative)

Component	Qty / size
Hot NVMe model (+ landing)	3–6 nodes/appliance, 150–300 TB usable
Archive object (EC)	$\geq$ 750 TB usable (path to 2 PB)
400GbE to leaves	2–4 ports per storage node/pair
Ingest brokers (if used)	3 VMs/nodes
OOB	All BMCs $\rightarrow$ 7010TX-48

7.11 Failure and Lifecycle

Event	Behavior
Hot FS node loss	Models still load; slower start
Archive node loss	No loss of committed objects
Bus backlog	Alert; do not block tokens unless policy=strict
Fabric leaf loss	ECMP dual-home
Day-366	Lifecycle delete or legal-hold
Hot FS outage (steady)	Live QPS OK if weights loaded; new starts fail
Archive outage	Live QPS OK; gap alarms; buffer replay

7.12 Summary Integration Map

Plane	System	Storage role
Data fabric	7060DX5	Models + archive + research scans
OOB	7010TX-48	BMCs only
Hot FS	CSI RWX	Checkpoints
Object lake	S3 API	1-year inference + artifacts
Bus	Kafka/Redpanda	Async capture
Local NVMe	GPU server	Cache / emit buffer
GPU HBM	B200	Steady-state weights

8 Kubernetes Architecture

8.1 Management Nodes

Quantity: 3 (HA control plane)

Component	Spec
CPU	32 cores
RAM	256 GB
Storage	NVMe

8.2 GPU Workers

worker01 ... worker08
each: 8× B200

8.3 Software Stack

```
Kubernetes
├─ NVIDIA GPU Operator
│   └─ CUDA
│       └─ TensorRT-LLM / vLLM
│           └─ KServe
│               └─ Ray (research / batch)
```

Layer	Role
Kubernetes	Scheduling, HA, multi-tenancy
GPU Operator	Driver, device plugin, DCGM, MIG
vLLM / TensorRT-LLM	High-performance serving
KServe	CRDs, canary, autoscaling hooks
Ray	Research jobs, eval, offline batch

9 Inference Request Flow

User → API Gateway → AuthN/Z → Inference Scheduler
→ Model Server → GPU Cluster → Response Tokens
↘ async InferenceRecord → 1-year archive

9.1 Serving Requirements

Attribute	Target direction
Latency	Interactive (TTFT + TPOT tuned)
Context	Up to 32k (initial planning)
Concurrency	50 simultaneous users
Throughput	1,000 tokens/sec aggregate

50 concurrent × 20 tokens/sec/user ≈ 1,000 tokens/sec

A 64× B200 system is in the correct range for this profile on a large MoE, subject to model/kernel tuning.

10 Power

10.1 Per Rack (32 GPUs / 4 servers)

Component	Watts
4 GPU servers	40 kW
Network	2 kW
Storage share	3 kW
Management share	1 kW
Rack total	46 kW

10.2 Facility

Item	Spec
Baseline racks	4: GPU-1, GPU-2, BOOT, STOR
Cluster IT load	100–115 kW
Available facility	130–150 kW recommended
GPU racks	43–46 kW each · 415 VAC A/B
BOOT	4–8 kW (CP, bootstrap, spines)
STOR	5–15 kW (hot FS + year archive)
STOR-2	Optional when archive > 1.5–2 PB
Feeds	A/B redundant all racks

Storage racks

Day-1 needs **one** STOR rack for hot models + ≥750 TB archive. Second storage rack only for capacity/isolation growth — not required at 64-GPU scale.

11 Cooling

Tier	Approach
Minimum	Hot-aisle containment + high-airflow CRAC
Preferred	Rear-door heat exchangers
Future	Direct liquid cooling (DLC)

Plan layout so DLC retrofit is possible.

12 Monitoring and Observability

12.1 GPU

NVIDIA DCGM · utilization · HBM · ECC · NVLink health · power / thermal

12.2 Network

Interface errors · ECN marks · drops · RoCE congestion · fabric latency/loss · LANS

12.3 Application

Tokens/sec · TTFT · TPOT · queue depth · GPU / KV-cache occupancy · errors/timeouts · **archive lag & gap alarms**

13 High Availability

Domain	Approach
Control plane	3-node Kubernetes HA
Fabric	Dual NIC, dual leaf (DX5-32), dual spine (DX5-64) ECMP + DLB
Power	A/B feeds
Model serving	Multi-replica where possible; KServe rolling updates
Storage	Hot FS repl/EC + archive object EC; 365d retention

Align failure domains with racks and leaf pairs so a single leaf/rack loss does not take the full model offline if EP groups are placed carefully.

14 Expandability

Path	Notes
+1 rack (32 GPUs)	Add leaf capacity, extend BGP EVPN, join workers
+server within rack	Power/cooling headroom first
Storage growth	Scale hot FS + object lake independently
Multi-model	Partition expert groups / namespaces by team

15 Approximate Cost

Category	Estimate
64× B200 GPUs	\$4-6M
GPU servers	\$1-1.5M
Network fabric	\$1-1.5M
Storage (hot + archive)	\$0.5-1.5M+
Rack / power / cooling	\$500k
Total	\$7-11M+

Order-of-magnitude planning only; obtain vendor quotes. Archive lake size drives storage variance.

16 Final Design Recommendation

Layer	Choice
Compute	64× B200 / 8 nodes / 2 GPU racks
Facility	4 racks: GPU-1/2 + BOOT + STOR
Network	8×400G/node rails: 8×DX5-32 + 2×DX5-64S; 7010TX-48 OOB
Storage	Hot NVMe models + ≥750 TB 1-year inference lake
Platform	Kubernetes + GPU Operator + vLLM/TRT-LLM + KServe + Ray
Ops	DCGM + EOS/LANZ + token SLOs + archive lag

This is the scale at which the system behaves like a real AI supercomputer while remaining operable by an enterprise infrastructure team.

17 Document Roadmap

1. `docs/bom.md` — bill of materials and SKUs
2. `docs/network.md` — underlay, RoCE, QoS, IP plan
3. `docs/k8s.md` — cluster bootstrap, GPU Operator, serving
4. `docs/storage.md` — model repo, archive schema, performance
5. `docs/facility.md` — rack elevations, power, cooling
6. `docs/slo.md` — latency/throughput SLOs and capacity model

18 Revision History

Ver	Date	Notes
0.1	2026-07-18	Initial specification
0.2	2026-07-18	Network: Arista 7060DX5 (DX5-32 leaf, DX5-64S spine)
0.3	2026-07-18	Lossless $\neq$ equal leaf/spine; square Clos Option B
0.4	2026-07-18	OOB: Arista 7010TX-48 (2×, 1 per rack)
0.5	2026-07-18	Storage integration: fabric, CSI/KServe, tiers
0.6	2026-07-18	1-year inference archive lake ($\geq$ 750 TB, async)